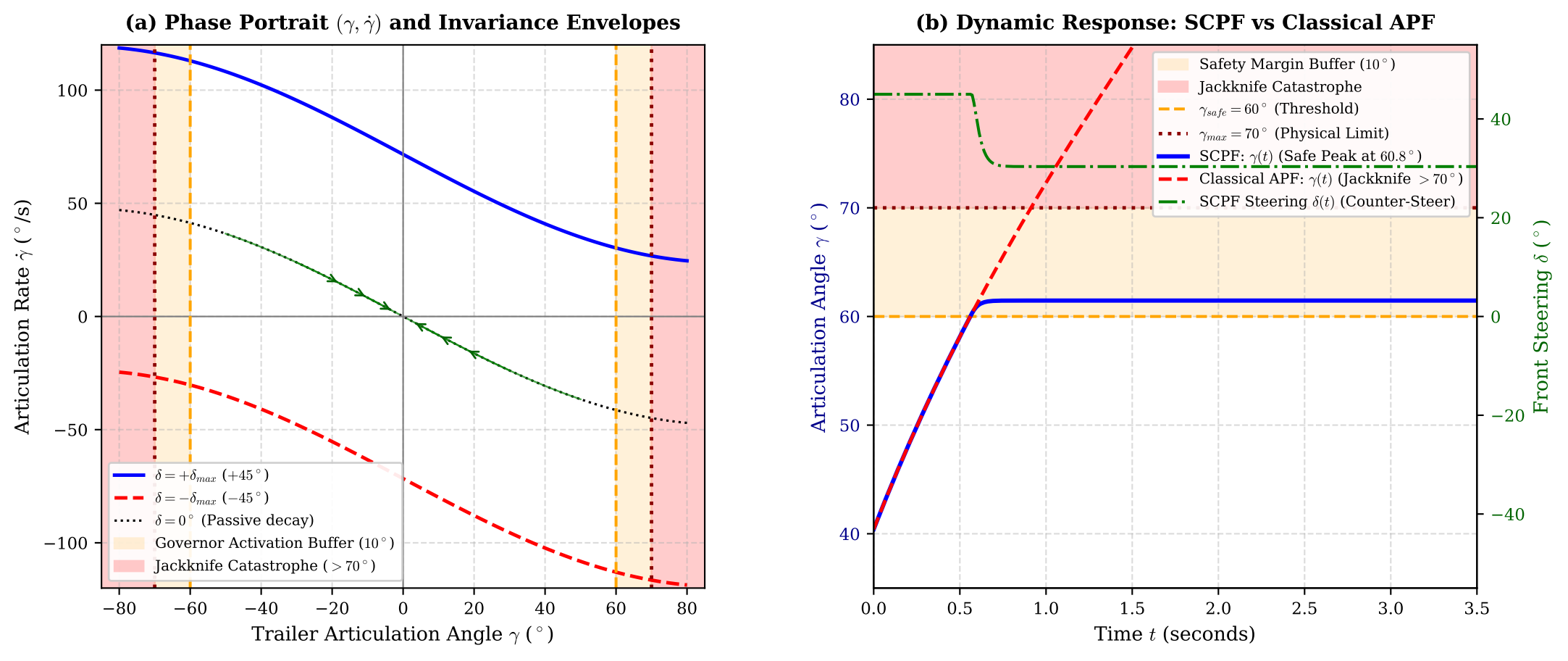
(a) Phase Portrait $(\gamma, \dot{\gamma})$ and Invariance Envelopes
Articulation Rate $\dot{\gamma}$ (°/s)
Trailer Articulation Angle γ (°)
$\delta = +\delta_{max}$ (+45°)
$\delta = -\delta_{max}$ (−45°)
$\delta = 0°$ (Passive decay)
Governor Activation Buffer (10°)
Jackknife Catastrophe (>70°)
(b) Dynamic Response: SCPF vs Classical APF
Articulation Angle γ (°)
Front Steering δ (°)
Time t (seconds)
Safety Margin Buffer (10°)
Jackknife Catastrophe
$\gamma_{safe} = 60°$ (Threshold)
$\gamma_{max} = 70°$ (Physical Limit)
SCPF: $\gamma(t)$ (Safe Peak at 60.8°)
Classical APF: $\gamma(t)$ (Jackknife >70°)
SCPF Steering $\delta(t)$ (Counter-Steer)